MAITRI SYSTEM ARCHITECTURE & ENGINEERING REPORT

Multimodal AI Assistant for Psychological & Physical Well-Being of Astronauts · ISRO BAS

Problem ID: 25175	**Target Station:** Bhartiya Antariksh Station (BAS)
Author / Lead: MAITRI Engineering Team	**Target Orbit:** 410 km Low Earth Orbit (LEO)
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Security Classification: Restricted / ISRO DOS	**Verification Status:** 9/9 Automated Tests Passing (100%)

1. Executive Summary & Core Mission Objectives

MAITRI is an onboard, edge-native multimodal artificial intelligence companion engineered to preserve the mental, emotional, and physiological health of astronauts during long-duration orbital missions. Operating under the extreme constraints of spaceflight—microgravity cephalad fluid shifts, circadian desynchrony, communication delays, and deep isolation—MAITRI delivers non-intrusive, continuous biometric surveillance with situational psychological intervention.

Core Architectural Tenets:
• **100% Offline-First Edge Execution:** Operates completely without cloud dependencies; no visual or raw audio data leaves the spacecraft.
• **Multimodal Tri-Channel Perception:** Synchronizes computer vision (Facial Action Units), acoustic speech prosody (F0 pitch / vocal tension), and linguistic sentiment analysis.
• **Clinical Decision Support:** Calculates a deterministic 4-tier well-being index (0–100 scale) with cross-modal discordance detection to flag masked distress.
• **Strict Role-Based Access Control (RBAC):** Defends crew psychological privacy by partitioning astronaut-facing interactive companionship from ground flight surgeon diagnostic triage.

2. Subsystem Architecture & Implemented Features

2.1 AIML Multimodal Perception Pipeline (/AIML)

• **Facial Emotion Recognition (FER) Module:** Implements facial biomarker analysis compatible with OpenCV 5.0+. Features multi-spectral skin chrominance segmentation (YCrCb + HSV) to locate face contours regardless of cabin lighting, computing Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), smile curvature (AU12), and brow furrow tension (AU04). Evaluates blinks/min, yawns/min, and PERCLOS (Percentage of Eye Closure).
• **Speech Emotion Recognition (SER) Module:** Evaluates raw PCM audio streams via time-domain autocorrelation to extract fundamental frequency (F0 pitch in Hz), Root Mean Square (RMS) energy, and vocal tract muscle tension without cloud dependencies.
• **Linguistic Sentiment Engine:** Evaluates speech transcripts for cognitive load, emotional valence, and crisis cues ('panic', 'cannot breathe', 'failing').
• **Late Attention Fusion & Discordance Detector:** Computes dynamic attention weights (alpha, beta, gamma) across modalities. Identifies 'smiling depression' or masked distress when facial valence appears positive but vocal tension is critically elevated.
• **Deterministic Well-Being Risk Evaluator:** Maps fused states into 4 clinical tiers: Tier 0 (Nominal, 0-30), Tier 1 (Mild Fatigue, 31-50), Tier 2 (Moderate Distress, 51-70), Tier 3 (Critical Emergency, 71-100).
• **Cognitive Conversational Companion AI:** Powered by an integrated 42-topic space operations dataset with semantic search and an optional 1-click connector to Google Gemini 1.5 Flash.

2.2 Backend & Data Storage Subsystem (/Backend_DB)

• **FastAPI Server Engine:** High-throughput asynchronous REST endpoints and sub-50ms latency WebSocket streams (/ws/telemetry).
• **Database Management:** Thread-safe SQLite connection manager with automated schema migration (maitri_session.db). Logs time-series telemetry, alerts, and intervention tracking.
• **Intervention Protocol Dispatcher:** Triggers autonomous clinical actions including Tactical Box Breathing (INT-BREATHE-01), Circadian Lighting Adjustment (INT-FATIGUE-04), Earth Longing Uplinks (INT-EARTH-05), and Ground Medical Triage (INT-GROUND-02).
• **Report & Export Generators:** Generates multi-page vector PDF flight surgeon summaries (ReportLab), raw JSON telemetry dumps, and high-resolution JPG visual health passports.

2.3 Security & Role-Based Access Control (/Security_API)

• **Dual Role Hierarchy:** Enforces separation between ASTRONAUT (private self-care, breathing guides, mission checklists) and ADMIN (Flight Surgeon telemetry review, crew longitudinal triage, ground alert triggers).
• **Route Guards:** FastAPI dependencies enforce HTTP 403 Forbidden on administrative endpoints (/api/admin/*) if accessed by astronaut tokens.
• **Security Shield Middleware:** Masks sensitive internal stack traces, injects X-Content-Type-Options, X-Frame-Options, and CSP security headers.

2.4 Aerospace Web Interface (/maitri/web)

• **Executive Aerospace Top Navigation:** 60px height bar with dynamic role badges, crew callsign indicator, 1-click role switcher, AI Brain connector pill, and export triggers.
• **Biometric Hardware Lifecycle Banner:** Live status indicator reflecting Ready, Permission Granted, Blocked in Browser (with unblocking instructions), or Device Unavailable.
• **Dual-Layer Live Video & HUD Viewport:** Native 60 FPS camera streaming with an overlaid transparent canvas rendering green cybernetic reticles, eye/mouth landmarks, real-time EAR/MAR tags, and PERCLOS metrics.
• **Acoustic Waveform Canvas:** Autocorrelation pitch oscilloscope that displays live vocal frequencies and cleanly flatlines to a calm baseline when muted.
• **Emotional Valence 24H Timeline:** Real-time SVG timeline color-coded across Positive, Neutral, and Negative zones, throttled to prevent frame-flooding.
• **AI Brain Settings Modal:** 1-click interface to connect Google Gemini 1.5 Flash via free API keys.

2.5 DevOps & CI/CD Pipeline (/DevOps & /.github)

• **Automated Test Suite (run_tests.py):** Validates 9 verification points (system health, astronaut auth, flight surgeon auth, RBAC guard HTTP 403, admin access HTTP 200, simulation pipeline, companion AI, vector PDF export, and JSON/JPG export).
• **GitHub Actions CI Pipeline (.github/workflows/ci.yml):** Runs on ubuntu-latest, spins up server background daemons, executes automated tests, and builds Docker containers.
• **Production Dockerfile:** Multi-stage container setup with system audio/video dependencies (libgl1, libsndfile1) and non-root security.

3. Technical Challenges & Problem-Solving Approaches

During development, several critical edge-case engineering bugs and platform limitations were diagnosed and resolved. The table below summarizes the exact problem, root cause, and production approach implemented:

Challenge / Bug	Root Cause	Architectural Solution Implemented
OpenCV 5.0 Missing CascadeClassifier Facial expression analysis failed to detect faces.	The pre-release opencv-python 5.0.0 build on Python 3.14 lacks legacy cv2.CascadeClassifier, causing AttributeError.	Re-engineered FacialEmotionModule using multi-spectral skin chrominance segmentation (YCrCb + HSV) and Otsu adaptive thresholding. Computes EAR/MAR and FACS Action Units without XML cascade files.
Camera Video Canvas Blackout Camera permission was active but frames were blank.	The <video> element had class='hidden' (display: none). Modern browsers pause hardware decoders on hidden elements, causing drawImage to capture 0x0 black frames.	Restructured the viewport into a dual-layer stack: active <video> as base layer with a transparent <canvas> overlaid on top. Yields full 60 FPS video and crystal-clear cybernetic HUD reticles.
Audio Microphone Mute Leakage Audio readings and waveforms continued scrolling when mic was toggled off.	Audio stream sharing kept microphone hardware open, and audio analysis loop lacked a hardware kill-switch.	Separated microphone acquisition from camera stream. Stopping mic now calls track.stop(), clears stream buffers, resets pitch to 0 Hz, and draws a flat baseline.
Timeline 4 FPS Flooding Timeline graph was constantly shifting rapidly 4 times every second.	Every idle camera frame was appending data points to the historical timeline SVG array.	Throttled timeline SVG regeneration so it only shifts on authentic state changes, scenario triggers, or 5-second sampling intervals.
Rigid / 'Dumb' AI Dialogue Offline AI companion repeated stiff default responses for novel questions.	Keyword lookup lacked domain depth and defaulted to a generic cop-out whenever a prompt fell outside templates.	1) Created space_qa_dataset.json with 42 extensive aerospace categories. 2) Built a multi-signal semantic search engine with NLP intent decomposition. 3) Added a 1-click UI connector for free Google Gemini 1.5 Flash.
RBAC Route Privacy Leak Astronauts could view private flight surgeon logs.	Lack of role enforcement on administrative REST endpoints.	Implemented FastAPI dependency guards (require_role(UserRole.ADMIN)) returning HTTP 403 Forbidden for unauthorized requests.

4. Backend API Endpoints & Route Reference

Endpoint	Method	Auth / RBAC	Function Description
/api/status	GET	Public	Returns system health, active station, and module status.
/api/auth/login	POST	Public	Issues signed session tokens for astronauts or surgeons.
/api/auth/switch-role	POST	Session	1-click role switcher between Astronaut and Admin for demo.
/api/process_frame	POST	Astronaut	Ingests base64 frame & audio, returns multimodal telemetry.
/ws/telemetry	WS	Astronaut	Sub-50ms bidirectional WebSocket stream for HUD telemetry.
/api/interact	POST	Astronaut	Conversational companion dialogue generation.
/api/settings/ai-key	POST	Admin/User	Configures and activates Google Gemini 1.5 Flash API key.
/api/settings/ai-status	GET	Public	Returns active LLM brain status (online vs offline).
/api/simulate/{scenario}	POST	Public	Injects flight scenarios (Nominal, Fatigue, Stress, Discordance).
/api/admin/triage	GET	Flight Surgeon	RBAC guarded flight surgeon triage console (HTTP 403 for crew).
/api/export/pdf	GET	Astronaut	Generates official vector PDF flight surgeon report.
/api/export/json	GET	Astronaut	Downloads raw structured telemetry history as JSON.
/api/export/jpg	GET	Astronaut	Generates high-resolution JPG visual health passport card.

5. Strategic Engineering Roadmap (Phase 2 & Beyond)

To advance MAITRI from a winning SIH demonstration prototype to flight-certified spacecraft hardware, the following engineering phases are recommended:

1. Embedded Edge Hardware Acceleration (TRL 5 -> 7):
• Port OpenCV FER and acoustic extraction pipelines to NVIDIA Jetson Orin Nano or Google Coral Edge TPU using TensorRT / ONNX Runtime to maintain <15ms latency on low-power DC bus.
• Benchmark thermal dissipation in pressurized microgravity environments where convective cooling is absent.

2. Wearable Biosensor Integration (BLE & Ant+):
• Interface with commercial aerospace smartwatches (Garmin D2 Mach 1, Empatica E4) for continuous photoplethysmography (PPG) heart rate variability (HRV), pulse oximetry (SpO_2), and galvanic skin response (GSR).
• Integrate autonomic HRV metrics directly into the Well-Being Index calculation.

3. Local Quantized Large Language Model (SLM Deployment):
• Package a 4-bit quantized Small Language Model (e.g. Llama-3.2-3B-Instruct or Phi-3.5-mini via llama.cpp / ONNX) directly into the spacecraft image to provide 100% generative conversational intelligence completely offline without external API keys.

4. Federated Multi-Crew Privacy Mesh:
• Deploy local biometric nodes in individual crew sleep quarters that run federated model updates without transmitting identifiable visual records across the spacecraft LAN.

5. Spacecraft Bus Telemetry Ingestion:
• Connect MAITRI directly to BAS spacecraft MIL-STD-1553B / SpaceWire data buses to ingest real-time cabin CO_2 partial pressure, acoustic decibels, and radiation dosimeter readings.